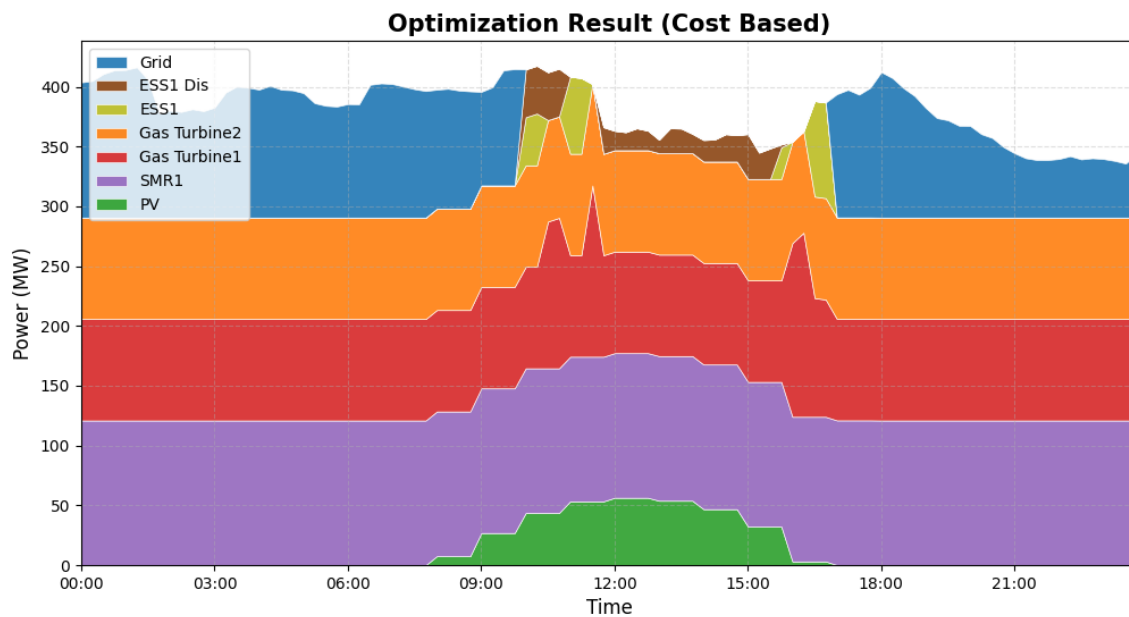


<Energy Dispatch Report>



1. Executive Summary

This report evaluates a microgrid portfolio composed of PV 56.3 MW, GT 340.0 MW, SMR 121.0 MW, and ESS 40.0 MW / 160.0 MWh over a 24-hour horizon. The optimization outcome indicates a portfolio centered on stable nuclear baseload, flexible gas support, daytime renewable injection, and targeted storage discharge during expensive periods.

Total cost reached 1,184,978,642 KRW, of which baseload generation cost accounted for 29,040,000 KRW (2.5%) and variable operating cost accounted for 1,077,111,976 KRW (90.9%). This indicates that short-term operational improvement is primarily determined by dispatch strategy, especially GT output management and on-peak grid substitution.

2. System Configuration Analysis

The asset portfolio combines a high-stability source, SMR (121.0 MW), with high-flexibility GT capacity (340.0 MW), medium-scale PV (56.3 MW), and an ESS sized at 40.0 MW / 160.0 MWh. From a strategic perspective, this mix is balanced toward reliability first, while still preserving opportunities for renewable integration and time-shifting.

The portfolio shows a deliberate separation of roles: SMR provides stable baseload, GT covers residual demand and ramping needs, PV offsets daytime energy, and ESS supports temporal balancing. This is a typical hybrid architecture for data center-class loads, where operational continuity and peak management are both critical.

3. Cost Structure Analysis

Cost decomposition shows that variable cost dominates the total economic outcome. Baseload generation cost contributes 2.5% of total cost, while variable cost contributes 90.9%. This confirms that the optimization problem is governed more by dispatch responsiveness than by fixed portfolio ownership.

In practical terms, this means cost efficiency depends on limiting GT-intensive and grid-dependent operation during high-price intervals. Even though SMR is a large-capacity asset, its cost contribution remains comparatively stable, while daily variability is driven by GT, PV availability, and ESS utilization.

4. Dispatch Strategy Analysis by Generation Source

Generation-source analysis confirms that each resource was assigned a differentiated operating role consistent

with its technical and economic characteristics.

- SMR: Average output 121.0 MW with capacity factor 100.0% over 24 hours. It functioned as the dominant baseload source with high stability and low short-term variability.
- GT: Average output 173.2 MW with capacity factor 50.9%. GT served as the flexible thermal resource, absorbing residual demand and supporting peak conditions.
- PV: Average output 15.3 MW, total energy 323.8 MWh, and capacity factor 24.0%. PV mainly contributed during daytime high-value periods, improving renewable penetration and lowering marginal energy cost.
- ESS: Average discharge 5.8 MW, total discharge 121.6 MWh, and discharge-based capacity factor 12.7%. ESS was used selectively for peak shaving and energy arbitrage, indicating a control strategy focused on value capture rather than continuous cycling.

5. TOU-Based Operation Strategy Analysis

- Off-Peak: SMR 121.0 MW | GT 169.7 MW | PV 0.8 MW | ESS 0.0 MW / 126.6 MWh
- Mid-Peak: SMR 121.0 MW | GT 169.7 MW | PV 3.9 MW | ESS 0.0 MW / 34.3 MWh
- On-Peak: SMR 121.0 MW | GT 180.2 MW | PV 41.3 MW | ESS 17.4 MW / 65.2 MWh

GT output increases from 169.7 MW in Off-Peak to 180.2 MW in On-Peak, a rise of 10.5 MW. This variation confirms that GT is the primary balancing asset under TOU-based cost signals.

During On-Peak periods, PV output rises to 41.3 MW and ESS discharge reaches 17.4 MW, producing a combined non-fossil / stored support effect of 58.7 MW. This reduces reliance on higher-cost thermal or imported power at the most valuable time of day.

The TOU strategy minimizes cost by shifting storage contribution and renewable value toward expensive hours while maintaining SMR as a stable base layer and using GT only as the adjustable balancing margin.

6. Overall Assessment and Recommendations

Overall assessment: the optimization result is strong in operational stability, clear resource role allocation, and effective On-Peak support from PV and ESS. However, variable cost remains dominant at 90.9% of total cost, which implies continued sensitivity to thermal dispatch and market price exposure.

- Recommendation 1: expand or re-optimize ESS operation. Current average discharge of 5.8 MW suggests storage is valuable but selectively used; increasing usable power or duration could further reduce GT and grid exposure during On-Peak hours.
- Recommendation 2: reduce GT dependency through additional low-marginal-cost supply or tighter commitment logic. GT remains the main flexible asset, so even modest reductions in peak GT output would materially improve variable cost.
- Recommendation 3: improve PV utilization through curtailment minimization, demand alignment, or coordinated ESS charging. This would increase daytime renewable capture and raise the effective value of the existing solar asset.
- Recommendation 4: evaluate scenario-based TOU sensitivity. Testing alternative tariff conditions and fuel cost assumptions would improve confidence in operating policy and investment prioritization.

7. Data Limitations and Assumptions

- The report assumes a 24-hour analysis horizon and interprets all average outputs over that period.
- Capacity factor values are calculated from average output or total discharge divided by rated capacity over 24 hours.
- Baseload generation cost is interpreted from the modeled low-variability generation block, while total cost and variable cost are taken directly from optimization output.
- External constraints such as maintenance schedules, reserve requirements, emissions limits, and forecast

uncertainty are not explicitly represented in this narrative unless reflected in the optimization result.

- Additional model note: Technical Report on Microgrid Energy System

Detailed Simulation Data (24h)

Time	Grid	PV	Gas Turbine	Gas Turbine	SMR1	ESS1	ESS1	Tot	Dif
00:00	113.0	0.0	84.8	84.8	121.0	0.0	0.0	403.7	40.0
00:15	114.2	0.0	84.8	84.8	121.0	0.0	0.0	404.9	40.0
00:30	120.0	0.0	84.8	84.8	121.0	0.0	0.0	410.7	40.0
00:45	123.2	0.0	84.8	84.8	121.0	0.0	0.0	413.9	40.0
01:00	123.5	0.0	84.8	84.8	121.0	0.0	0.0	414.2	40.0
01:15	125.7	0.0	84.8	84.8	121.0	0.0	0.0	416.4	40.0
01:30	115.0	0.0	84.8	84.8	121.0	0.0	0.0	405.7	29.5
01:45	87.0	0.0	84.8	84.8	121.0	0.0	0.0	377.7	0.0
02:00	89.4	0.0	84.8	84.8	121.0	0.0	0.0	380.1	0.0
02:15	88.3	0.0	84.8	84.8	121.0	0.0	0.0	379.0	0.0
02:30	90.6	0.0	84.8	84.8	121.0	0.0	0.0	381.3	0.0
02:45	89.0	0.0	84.8	84.8	121.0	0.0	0.0	379.7	0.0
03:00	92.0	0.0	84.8	84.8	121.0	0.0	0.0	382.7	0.0
03:15	104.5	0.0	84.8	84.8	121.0	0.0	0.0	395.2	0.0
03:30	109.6	0.0	84.8	84.8	121.0	0.0	0.0	400.3	0.0
03:45	108.6	0.0	84.8	84.8	121.0	0.0	0.0	399.3	0.0
04:00	107.0	0.0	84.8	84.8	121.0	0.0	0.0	397.7	0.0
04:15	110.2	0.0	84.8	84.8	121.0	0.0	0.0	400.9	0.0
04:30	106.9	0.0	84.8	84.8	121.0	0.0	0.0	397.6	0.0
04:45	106.5	0.0	84.8	84.8	121.0	0.0	0.0	397.2	0.0
05:00	104.1	0.0	84.8	84.8	121.0	0.0	0.0	394.8	0.0
05:15	95.7	0.0	84.8	84.8	121.0	0.0	0.0	386.4	0.0
05:30	93.5	0.0	84.8	84.8	121.0	0.0	0.0	384.2	0.0
05:45	92.8	0.0	84.8	84.8	121.0	0.0	0.0	383.5	0.0
06:00	94.9	0.0	84.8	84.8	121.0	0.0	0.0	385.6	0.0
06:15	94.7	0.0	84.8	84.8	121.0	0.0	0.0	385.4	0.0
06:30	111.0	0.0	84.8	84.8	121.0	0.0	0.0	401.7	0.0
06:45	112.2	0.0	84.8	84.8	121.0	0.0	0.0	402.9	0.0
07:00	111.7	0.0	84.8	84.8	121.0	0.0	0.0	402.4	0.0
07:15	109.5	0.0	84.8	84.8	121.0	0.0	0.0	400.2	0.0
07:30	107.4	0.0	84.8	84.8	121.0	0.0	0.0	398.1	0.0
07:45	105.8	0.0	84.8	84.8	121.0	0.0	0.0	396.5	0.0
08:00	99.4	7.5	84.8	84.8	121.0	0.0	0.0	390.1	0.0
08:15	100.2	7.5	84.8	84.8	121.0	0.0	0.0	390.9	0.0
08:30	98.7	7.5	84.8	84.8	121.0	0.0	0.0	389.4	0.0
08:45	98.0	7.5	84.8	84.8	121.0	0.0	0.0	388.7	0.0
09:00	78.3	26.9	84.8	84.8	121.0	0.0	0.0	369.0	0.0
09:15	82.1	26.9	84.8	84.8	121.0	0.0	0.0	372.8	0.0
09:30	96.3	26.9	84.8	84.8	121.0	0.0	0.0	387.0	0.0
09:45	97.3	26.9	84.8	84.8	121.0	0.0	0.0	388.0	0.0
10:00	0.0	43.6	84.8	84.8	121.0	40.3	40.0	371.0	0.0
10:15	0.0	43.6	84.8	84.8	121.0	43.3	40.0	374.0	0.0
10:30	0.0	43.6	122.7	84.8	121.0	0.0	40.0	368.6	0.0
10:45	0.0	43.6	125.8	84.8	121.0	0.0	40.0	371.6	0.0
11:00	0.0	53.3	84.8	84.8	121.0	64.5	0.0	355.2	0.0
11:15	0.0	53.3	84.8	84.8	121.0	63.4	0.0	354.1	0.0
11:30	0.0	53.3	143.2	84.8	121.0	0.0	0.0	349.1	0.0
11:45	0.0	53.3	84.8	84.8	121.0	0.0	22.2	312.9	0.0

Time	Grid	PV	Gas Turbine	Gas Turbine	SMR1	ESS1	ESS1	Tot	Dif
12:00	0.0	56.3	84.8	84.8	121.0	0.0	16.2	306.9	0.0
12:15	0.0	56.3	84.8	84.8	121.0	0.0	15.1	305.8	0.0
12:30	0.0	56.3	84.8	84.8	121.0	0.0	18.4	309.1	0.0
12:45	0.0	56.3	84.8	84.8	121.0	0.0	16.4	307.1	0.0
13:00	0.0	53.9	84.8	84.8	121.0	0.0	11.3	302.0	0.0
13:15	0.0	53.9	84.8	84.8	121.0	0.0	20.8	311.5	0.0
13:30	0.0	53.9	84.8	84.8	121.0	0.0	20.6	311.3	0.0
13:45	0.0	53.9	84.8	84.8	121.0	0.0	16.2	306.9	0.0
14:00	0.0	46.8	84.8	84.8	121.0	0.0	18.1	308.8	0.0
14:15	0.0	46.8	84.8	84.8	121.0	0.0	18.4	309.1	0.0
14:30	0.0	46.8	84.8	84.8	121.0	0.0	22.9	313.6	0.0
14:45	0.0	46.8	84.8	84.8	121.0	0.0	22.2	312.9	0.0
15:00	0.0	32.3	84.8	84.8	121.0	0.0	37.6	328.3	0.0
15:15	0.0	32.3	84.8	84.8	121.0	0.0	22.1	312.8	0.0
15:30	0.0	32.3	84.8	84.8	121.0	0.0	25.6	316.3	0.0
15:45	0.0	32.3	84.8	84.8	121.0	26.5	2.4	319.7	0.0
16:00	0.0	3.1	145.3	84.8	121.0	0.0	0.0	351.2	0.0
16:15	0.0	3.1	154.0	84.8	121.0	0.0	0.0	359.8	0.0
16:30	0.0	3.1	99.3	84.8	121.0	80.0	0.0	385.2	0.0
16:45	0.0	3.1	98.0	84.8	121.0	80.0	0.0	383.9	0.0
17:00	103.0	0.1	84.8	84.8	121.0	0.0	0.0	393.7	0.0
17:15	106.7	0.1	84.8	84.8	121.0	0.0	0.0	397.4	0.0
17:30	102.7	0.1	84.8	84.8	121.0	0.0	0.0	393.4	0.0
17:45	108.4	0.1	84.8	84.8	121.0	0.0	0.0	399.1	0.0
18:00	121.6	0.0	84.8	84.8	121.0	0.0	0.0	412.3	0.0
18:15	116.7	0.0	84.8	84.8	121.0	0.0	0.0	407.4	0.0
18:30	108.2	0.0	84.8	84.8	121.0	0.0	0.0	398.9	0.0
18:45	101.7	0.0	84.8	84.8	121.0	0.0	0.0	392.4	0.0
19:00	91.6	0.0	84.8	84.8	121.0	0.0	0.0	382.3	0.0
19:15	83.2	0.0	84.8	84.8	121.0	0.0	0.0	373.9	0.0
19:30	81.4	0.0	84.8	84.8	121.0	0.0	0.0	372.1	0.0
19:45	76.8	0.0	84.8	84.8	121.0	0.0	0.0	367.5	0.0
20:00	76.6	0.0	84.8	84.8	121.0	0.0	0.0	367.3	0.0
20:15	69.8	0.0	84.8	84.8	121.0	0.0	0.0	360.5	0.0
20:30	66.5	0.0	84.8	84.8	121.0	0.0	0.0	357.2	0.0
20:45	59.0	0.0	84.8	84.8	121.0	0.0	0.0	349.7	0.0
21:00	53.8	0.0	84.8	84.8	121.0	0.0	0.0	344.5	0.0
21:15	49.6	0.0	84.8	84.8	121.0	0.0	0.0	340.3	0.0
21:30	48.1	0.0	84.8	84.8	121.0	0.0	0.0	338.8	0.0
21:45	48.1	0.0	84.8	84.8	121.0	0.0	0.0	338.8	0.0
22:00	49.0	0.0	84.8	84.8	121.0	0.0	0.0	339.7	0.0
22:15	51.3	0.0	84.8	84.8	121.0	0.0	0.0	342.0	0.0
22:30	48.7	0.0	84.8	84.8	121.0	0.0	0.0	339.4	0.0
22:45	49.6	0.0	84.8	84.8	121.0	0.0	0.0	340.3	0.0
23:00	49.0	0.0	84.8	84.8	121.0	0.0	0.0	339.7	0.0
23:15	47.4	0.0	84.8	84.8	121.0	0.0	0.0	338.1	0.0
23:30	45.2	0.0	84.8	84.8	121.0	0.0	0.0	335.9	0.0
23:45	53.1	0.0	84.8	84.8	121.0	0.0	0.0	343.8	0.0